

The National Higher School of Artificial Intelligence
Introduction to Artificial Intelligence
Midterm Exam 18/05/2023
Duration: 1 hour 45 minutes

- 1. ANSWER ON THESE EXERCISE STATEMENT PAGES. IF YOU DO NOT HAVE ENOUGH, SPACE WRITE ON THE BACK OF THE SAME PAGE.**
- 2. DO NOT USE ANY RED OR PINK (OR RELATED) COLOUR IN YOUR ANSWERS!**
- 3. WRITE NEATLY SO AS NOT TO BE PENALISED!**

Exercise 1: 3 points

Answer and give a brief explanation to the following questions.

1. A definition of Artificial Intelligence is that it is “the study of how to make computers do things at which, at the moment, people are better”. Which school of AI does this definition point to? Explain.

2. Another definition of Artificial Intelligence is that it is “a field of study that tries to explain and emulate intelligent behaviour in terms of computational processes”. Which school of AI does this definition point to? Explain.

3. What is cognitive science?

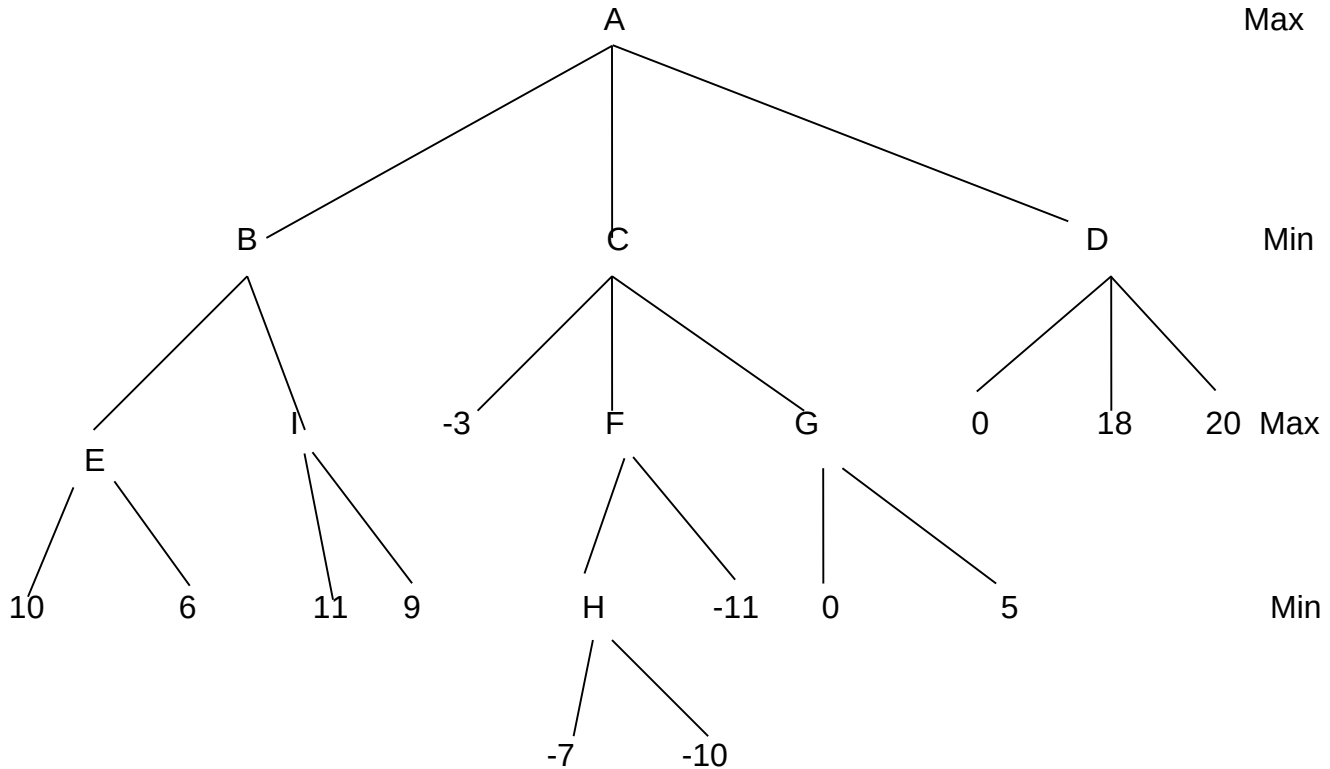
4. Suppose that we want to represent a node of a search tree as a record. What information would be needed in such a node to implement the general search algorithm?

5. Suppose we are using a path cost function with some search technique. Is it possible that the cost of a path decreases going from the root downward? If so, say when. Could you give an example of a problem where this would happen?

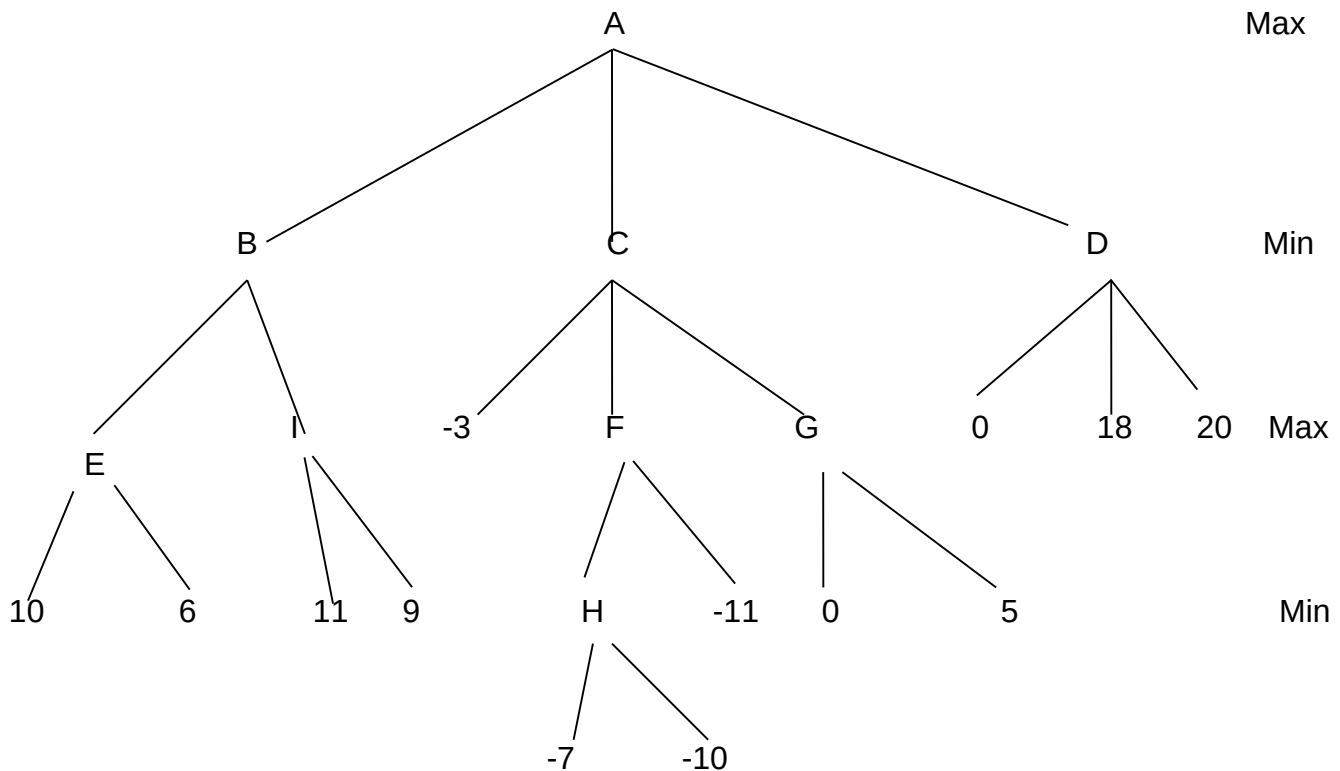
Exercise 2: 4 points

Consider the following 4-ply game.

1. Pass the values from the leaf nodes back up reflecting the work of the minimax procedure.



2. Use the minimax procedure with the alpha-beta cutoffs, showing ALL the values as per the algorithm and clearly show which part(s) would be pruned if at all.



Exercise 3: 7 points

Consider the 8-puzzle problem where we want to move from the configuration

2	8	3
1	6	4
7		5

to the configuration

1	2	3
8		4
7	6	5

In our particular case, we do not allow the **generation** of a state that has already been generated higher up, on the **same path** from the root to the current state.

In any search tree in the following questions, label each branch with the action that gets performed and USE THE APPLICABLE ACTIONS ALWAYS IN THE SAME ORDER.

1. Define the problem. (Make sure you explain your notation and conventions very clearly.)
2. Draw on the back of this page in the landscape orientation (i.e. using the width of the page) the tree that would be produced by breadth-first search, supposing that we set a limit to depth 3.

3. Define a heuristic function that could be used with A* search. Draw on the back of this page the tree produced by A* search using your heuristic function. Make sure you indicate above each state in the tree the relevant cost.
4. Analyse the tree generated as a result of questions 2 and 3 to define an evaluation function that would be suitable for Hill Climbing. State it and use it to solve the problem (with the Hill Climbing strategy).

5. Briefly say what you can conclude comparing the results you got at questions 3, 4, and 5.

Exercise 4: 6 points

Consider the following cryptarithmic problem where all the letters take values between 0 and 9 and each letter must be different from all others.

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  MEN
  ARE
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 WISE
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1. State the CSP problem precisely and define the domains entirely.
2. Using the studied heuristics appropriately and explaining which one is selected at any point, or which arithmetic/propagation is done, show on a search tree (making clear any backtracking that occurs) whether the problem has a solution or not. If it does have one, give it. Each node in the search tree will show all the applicable constraints at that point. (In order not to get lost/confused, make sure you apply the heuristics systematically as explained in the lectures.) Do not use any intelligent backtracking.

